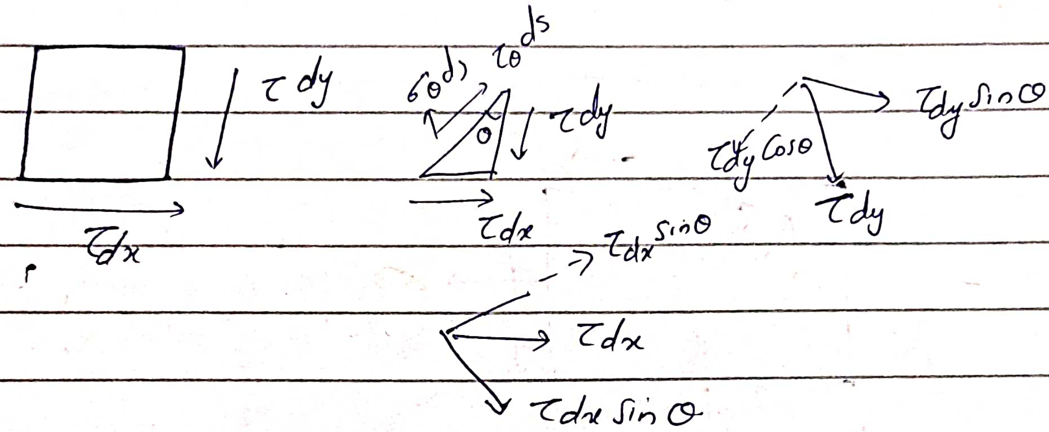
17/01/24

we Derived Eqn to find Normal & sheer stress in (1) uniaxial and (2) Bi-axial.

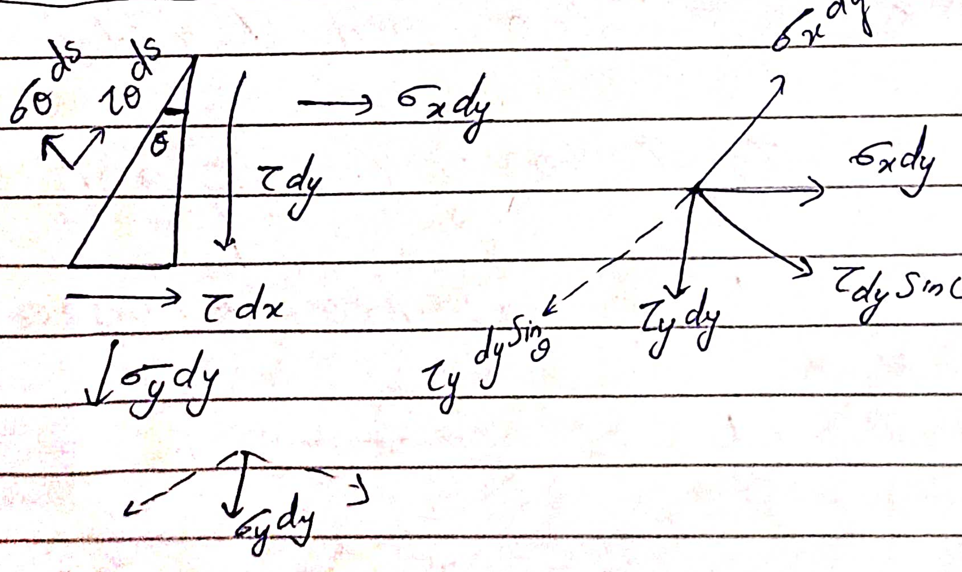
(3) Pure sheer



By Doing force balance,

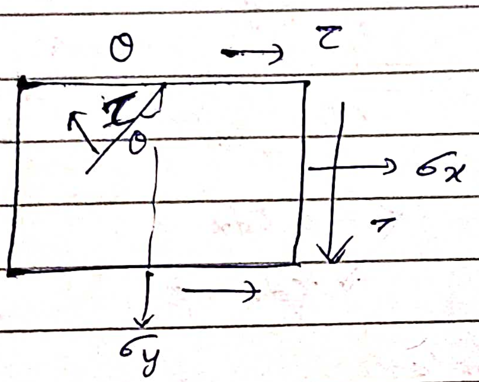
$$\sigma_\theta = \tau \sin 2\theta$$
$$\tau_\theta = \tau \cos 2\theta$$

(4) Bi-axial with sheer stress



$$\sigma_\theta = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau \sin 2\theta.$$

$$\tau_\theta = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau \cos 2\theta.$$



• Planes in which there is ~~only~~ no shear stress. only ~~real~~ normal stress.

if ~~θ = θ~~; $\tau_\theta = 0$ → no shear stress

$$\tau_\theta = 0 = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau \cos 2\theta$$

$$\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta = \tau \cos 2\theta.$$

$$\frac{1}{2}(\sigma_x - \sigma_y)\tan 2\theta = \tau$$

$$(\sigma_x - \sigma_y)\tan 2\theta = 2\tau$$

at $\tau_\theta = 0$. $\boxed{\tan 2\theta = \frac{2\tau}{(\sigma_x - \sigma_y)}}$

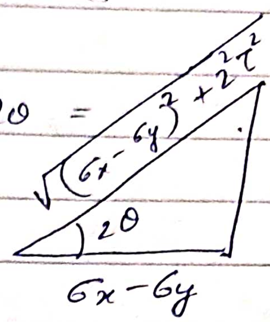
They are known as principle planes.

And these stresses are called principle stress.

Lets find the σ values at a plane.

$$\sin 2\theta = \overset{\text{direction}}{\pm} \frac{2\tau}{\sqrt{(\sigma_x-\sigma_y)^2+(2\tau)^2}}$$

$\tan 2\theta =$



$$\cos 2\theta = \pm \frac{\sigma_x-\sigma_y}{\sqrt{(\sigma_x-\sigma_y)^2+(2\tau)^2}}$$

This is done to replace $\sin 2\theta$ & $\cos 2\theta$

Substitute in eqn $\sigma_\theta = \frac{1}{2}(\sigma_x+\sigma_y) + \frac{1}{2}(\sigma_x-\sigma_y).\cos 2\theta + \tau \sin 2\theta$

ie, $$\frac{1}{2}(\sigma_x+\sigma_y) \pm \frac{1}{2}(\sigma_x-\sigma_y)\cdot\frac{(\sigma_x-\sigma_y)}{\sqrt{(\sigma_x-\sigma_y)^2+(2\tau)^2}} \pm \tau\cdot\frac{2\tau}{\sqrt{(\sigma_x-\sigma_y)^2+(2\tau)^2}}$$

Simplyfing...

$$= \frac{1}{2}(\sigma_x+\sigma_y) \pm \frac{1}{2}\frac{(\sigma_x-\sigma_y)^2}{\sqrt{(\sigma_x-\sigma_y)^2 \cancel{+(2\theta)^2} + 4\tau^2}} + \frac{2\tau^2}{\sqrt{(\sigma_x-\sigma_y)^2+4\tau^2}} \times \frac{1}{2}$$

$$= \frac{1}{2}(\sigma_x+\sigma_y) \pm \frac{1}{2}$$

$$\boxed{\sigma_{1,2} = \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}}$$

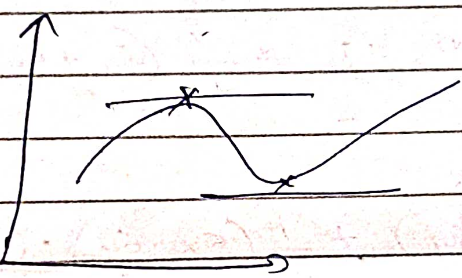
These are principle stresses σ_1 & σ_2

and $\tan 2\theta = \dfrac{2\tau}{(\sigma_x - \sigma_y)}$

ii) To find maximum sheer stress Case

$y = f(x)$

$f' = \dfrac{f_2 - f_1}{x_2 - x_1}$

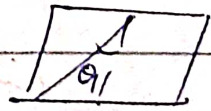


$$\frac{d\tau_\theta}{d\theta} = 0$$

$\dfrac{dy}{dx} = f' = 0$ $\tan 2\theta =$

$$\frac{d\tau_\theta}{d\theta} = 0 \Rightarrow \frac{-1}{2}(\sigma_x - \sigma_y)\, 2\cos 2\theta - 2\tau \sin\theta.$$

$$\tan 2\theta = -\left(\frac{\sigma_x - \sigma_y}{2\tau}\right)$$



~~There is only Normal~~

Maximum sheer stress value will lie in the

$T_{max} =$

Principle stress

$$\sigma_{1,2} = \frac{1}{2}(\sigma_x - \sigma_y) \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

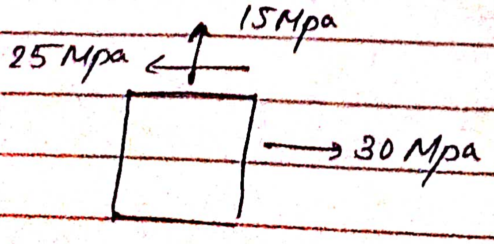
Principle stresses σ_1 σ_2 $\quad \tan 2\theta = \dfrac{2\tau}{(\sigma_x - \sigma_y)}$

Max shear stress

$$\tan 2\theta = -\frac{(\sigma_x - \sigma_y)}{2\tau}$$

$$\tau_{max} = \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

1) The stresses on two perpendicular planes through a point in a body are 30 Mpa and 15 Mpa Both tensile, along with ~~that~~ a shear stress of 25 Mpa. Find ① Magnitude & Direction of principle stresses
② plane of maximum shear stress
③ Normal & shear stresses on the planes of maximum shearing stress.



$\cancel{\sigma_z}$ $\left.\begin{matrix}\sigma_x \\ \sigma_y \\ \tau\end{matrix}\right\}$ Find &

Substitute

$\sigma_x = 30$ Mpa

$\sigma_y = 15$ Mpa

$\tau = 25$ Mpa

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$= \frac{1}{2}(45) + \frac{1}{2}\sqrt{(15)^2 + 4 \times 25^2}$$

$$= \cancel{225} + \frac{1}{2}\sqrt{225 + 2500}$$

$$= \overset{22.5}{\cancel{225}} + 26.2$$

$$= \underline{\underline{48.6}} \text{ Mpa } (T)$$

$$\sigma_2 = -3.6 \text{ Mpa } (C)$$

$$\cancel{\theta =} \quad \tan 2\theta = \frac{2\tau}{(\sigma_x - \sigma_y)}$$

$$2\theta = \tan^{-1}\left(\frac{2\tau}{\sigma_x - \sigma_y}\right)$$

$$\theta = \left(\frac{\tan^{-1}\left(\frac{2\tau}{\sigma_x - \sigma_y}\right)}{2}\right)$$

$$= \underline{\underline{36.65^\circ}}$$

② $\tan 2\theta = \dfrac{-(\sigma_x - \sigma_y)}{2\tau}$

$\theta = -8.35^\circ$

(iii) $\sigma_\theta = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau \sin 2\theta.$

$\tau_\theta = \frac{-1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau \cos 2\theta.$

Then

$\sigma_{-8.35} = \frac{1}{2}(45) + \frac{1}{2}(15)\cos(-8.35) + 25 \sin 2\theta$

$= 22.5 + \quad + \quad 7.18 \quad - 7.18$

$= 22.5\,Mpa$

$\tau_{-8.35} = -\frac{1}{2}(15)\sin(2\times -8.35) + 25\cos(2\times -8.35)$

$= 2.155 + 23.94$

$= 26.1\,Mpa$